

Wenrui Huang

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EDUCATION

Brown University

PhD in Mathematics

- Advisor: Benoît Pausader

Providence, RI

Aug 2022 — May 2027 (expected)

University of Science and Technology of China

B.S. in Mathematics

- Advisor: Yong Wei
- Thesis: *Mean curvature flow with surgery and its applications*

Hefei, China

Sep 2018 — May 2022

RESEARCH INTERESTS

Analysis and partial differential equations, especially nonlinear dispersive PDEs and kinetic equations

PUBLICATIONS AND PREPRINTS

1. *Scattering map for the Vlasov-Poisson system with a repulsive harmonic potential*, with H. Kwon, preprint.
2. *Stability and instability of small BGK waves*, with D. Bian, E. Grenier, and B. Pausader, preprint.
3. *The Vlasov-Poisson system with a perfectly conducting wall: Convex domains*, with B. Pausader and M. Suzuki, preprint.
4. *Modified wave operators for the Hartree equation with repulsive Coulomb potential*. preprint
5. *Scattering of the Vlasov-Riesz system in three dimensions*, with H. Kwon, preprint

TALKS

- GLESPA Seminar, Brown University, April 2026
- Poster session in Workshop on Kinetic Theory and Fluids, University of Wisconsin-Madison, March 2025.
- Brown-Yale PDE Seminar, Brown University, April 2024
- GLESPA Seminar, Brown University, April 2024
- Brown-Yale PDE Seminar, Brown University, November 2023

TEACHING EXPERIENCE

Brown University

- Fall 2026 — Instructor, MATH
- Fall 2025 — TA, MATH 0190 Calculus II (Physics/Engineering)
- Spring 2024 — TA, MATH 0180 Calculus III
- Fall 2023 — TA, MATH 0180 Calculus III
- Spring 2023 — TA, MATH 0100 Calculus II

University of Science and Technology of China

- Spring 2022 — TA, Riemannian Geometry
- Fall 2021 — TA, Differential Geometry
- Spring 2021, Spring 2022 — TA, Topology

OUTREACH AND SERVICE
